QICT INDEPENDENT PHYSICAL-POLE EXECUTION
Mathematical specification and fail-closed HPC handoff - 19 October 2026
STATUS: BLOCKED_MISSING_INPUT
Source repository: github.com/sachamed/qict-code-data
Observed main commit: fd664503ca92d2d88007d68d6e3873f7df2beeb8
Canonical action SHA256: c22ff80fcb94204984a65debf7e5636ef1d4e6caad89cf8fd6934dfdb8f684b7

1. CANONICAL ACTION
The frozen source describes a compact U(1), SU(2), SU(3) discrete-time gauge transfer.
Its BCC overlap/domain-wall class uses the primitive phase phi_star = pi/63.
The six source order is L1,L2,L3,R1,R2,R3 and spatial boundaries are periodic.
The action hash is verified; the complete physical U-cone and contour are not.

2. SPECTRAL REQUIREMENTS
For full physical W and source isometry V, C_n = V* W^n V and C_0 = I_6.
Projected resolvent G(z) = V* (z-W)^(-1) V, initially outside unit spectrum.
Toeplitz positivity alone cannot identify a physical pole from finite moments.
A true excitation requires an isolated pole with a positive, nonzero residue.
Its location and weight must be stable under volume, buffer and regulator changes.

3. AVAILABLE EVIDENCE IS INCOMPLETE
The L24 N9 rank-six file is a physical precontraction, not full apply_U.
Kato diagnostic and nested geometry files are not four physical Kato densities.
C10 shell geometry is not a verified complete quantum CP/Kraus map.
Synthetic control moments cannot establish a mass.

4. PHYSICAL OBSERVABLES
No dark particle pole, residue, dispersion or mass in GeV is computed.
A local NPZ read diagnostic is not a physical HPC matvec benchmark.

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5. COMPLETE PHYSICAL COMPUTATION ORDER

Kato/Riesz convergence with tail and contour gap; complete quantum C10 maps;
boundary-to-path amplitudes; physical core recoupling; rank-six isometry;
full interacting U-cone plus adjoint; CN6 moments and error-certified Toeplitz;
independent dark-sector source; isolated pole with positive residue;
multi-volume and regulator limits; independent clock calibration; physical mass.

6. MANDATORY VALIDATION

For each Kato buffer rho = F F* / Tr(F F*) with rho >= 0, Tr(rho) = 1.
The protecting Riesz gap must be positive on the entire contour.
Boundary insertion and C10 maps need independent CP/normalization checks.
The source map must obey V*V = I_6 and the unitary pass norm/adjoint tests.
Moment errors and Toeplitz positivity require independent rigorous bounds.
A finite-volume root or a block-pencil fit is diagnostic only.

7. VOLUME / DEPTH CHECKS

Volumes L = 24,32,48,64,96,128; depths 9,13,21,29,45,61.
Compare extrapolations, omit-one-volume tests and regulator sensitivity.
Never calibrate against a target particle mass.

8. DARK-SECTOR IDENTIFIABILITY

If W = W_charged direct-sum W_dark, and V = (V_charged,0), charged
moments do not depend on the spectrum of W_dark.
A stable neutral source V_chi with physical overlap must be constructed.

9. INDEPENDENT POLICY

Freeze the author-designated Git commit and entire computing environment.
Record SHA256 hashes, positive evidence and negative test results.
Perform a blinded physical extraction only after all gates pass.

DECISION: BLOCKED_MISSING_INPUT.
No numerical dark-particle mass may be reported as a QICT prediction.